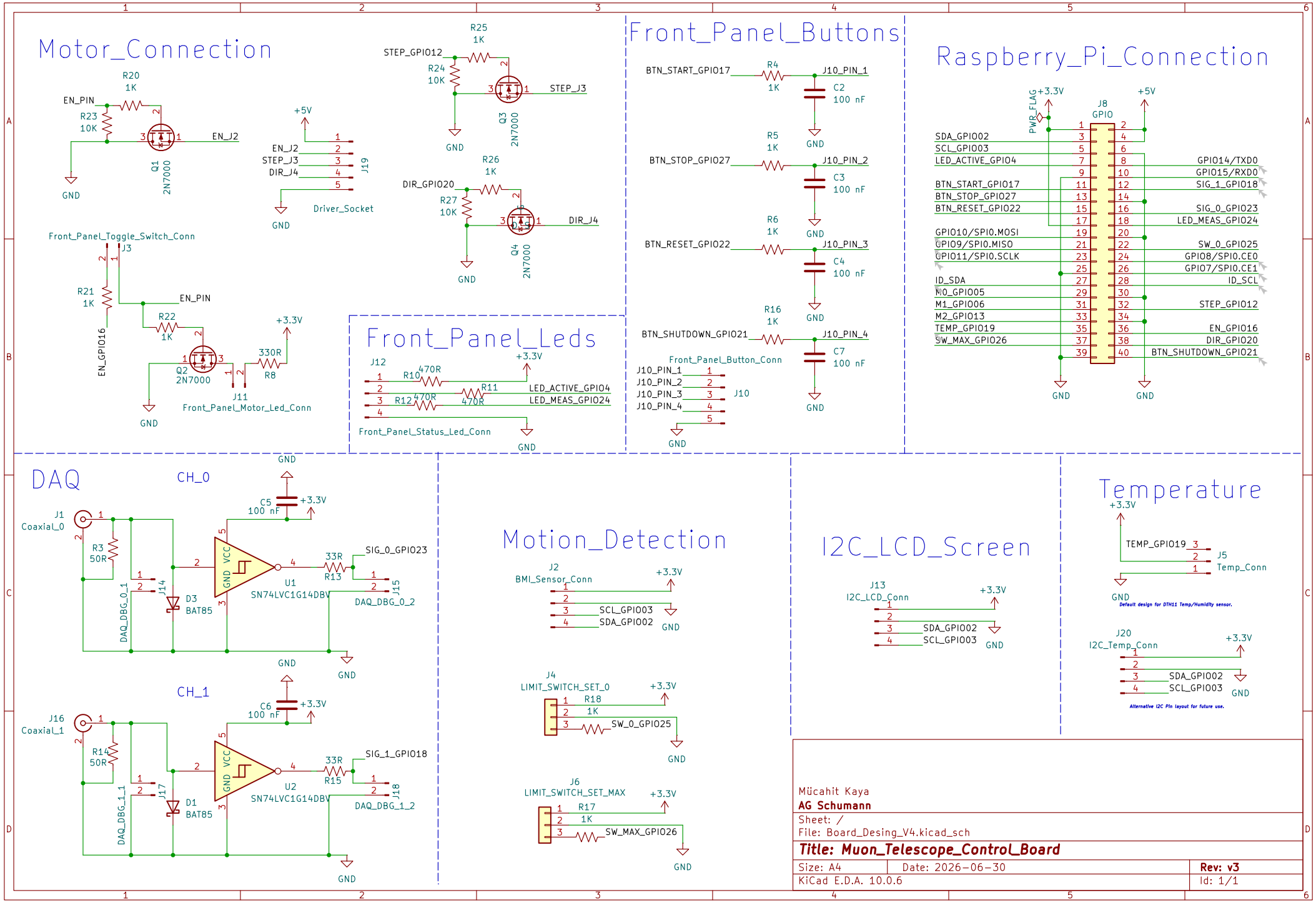

Motor_Connection
R20
1K
EN_PIN
R23
10K
Q1
2N7000
EN_J2
GND
+5V
EN_J2
STEP_J3
DIR_J4
J19
Driver_Socket
GND
R25
1K
STEP_GPIO12
R24
10K
Q3
2N7000
STEP_J3
GND
R26
1K
DIR_GPIO20
R27
10K
Q4
2N7000
DIR_J4
GND
Front_Panel_Toggle_Switch_Conn
J3
R21
1K
EN_PIN
EN_GPIO16
R22
1K
Q2
2N7000
330R
R8
+3.3V
J11
Front_Panel_Motor_Led_Conn
GND
Front_Panel_Leds
J12
+3.3V
R10
470R
R11
470R
LED_ACTIVE_GPIO04
R12
470R
LED_MEAS_GPIO24
Front_Panel_Status_Led_Conn
GND
Front_Panel_Buttons
BTN_START_GPIO17
R4
1K
J10_PIN_1
C2
100 nF
GND
BTN_STOP_GPIO27
R5
1K
J10_PIN_2
C3
100 nF
GND
BTN_RESET_GPIO22
R6
1K
J10_PIN_3
C4
100 nF
GND
BTN_SHUTDOWN_GPIO21
R16
1K
J10_PIN_4
C7
100 nF
GND
Front_Panel_Button_Conn
J10_PIN_1
J10_PIN_2
J10_PIN_3
J10_PIN_4
J10
GND
Raspberry_Pi_Connection
PWR_FLAG
+3.3V
+5V
J8
GPIO
SDA_GPIO02
SCL_GPIO03
LED_ACTIVE_GPIO04
BTN_START_GPIO17
BTN_STOP_GPIO27
BTN_RESET_GPIO22
GPIO10/SPI0.MOSI
GPIO9/SPI0.MISO
GPIO11/SPI0.SCLK
ID_SDA
M0_GPIO05
M1_GPIO06
M2_GPIO13
TEMP_GPIO19
SW_MAX_GPIO26
GPIO14/TXD0
GPIO15/RXD0
SIG_1_GPIO18
SIG_0_GPIO23
LED_MEAS_GPIO24
SW_0_GPIO25
GPIO8/SPI0.CE0
GPIO7/SPI0.CE1
ID_SCL
STEP_GPIO12
EN_GPIO16
DIR_GPIO20
BTN_SHUTDOWN_GPIO21
GND
GND
DAQ
CH_0
J1
Coaxial_0
R3
50R
C5
100 nF
+3.3V
GND
U1
SN74LVC1G14DBV
VCC
GND
33R
R13
SIG_0_GPIO23
J15
DAQ_DBG_0_2
J14
DAQ_DBG_0_1
D3
BAT85
GND
CH_1
J16
Coaxial_1
R14
50R
C6
100 nF
+3.3V
GND
U2
SN74LVC1G14DBV
33R
R15
SIG_1_GPIO18
J18
DAQ_DBG_1_2
J17
DAQ_DBG_1_1
D1
BAT85
GND
Motion_Detection
J2
BMI_Sensor_Conn
+3.3V
SCL_GPIO03
SDA_GPIO02
GND
J4
LIMIT_SWITCH_SET_0
R18
1K
+3.3V
SW_0_GPIO25
GND
J6
LIMIT_SWITCH_SET_MAX
R17
1K
+3.3V
SW_MAX_GPIO26
GND
I2C_LCD_Screen
J13
I2C_LCD_Conn
+3.3V
SDA_GPIO02
SCL_GPIO03
GND
Temperature
+3.3V
TEMP_GPIO19
J5
Temp_Conn
GND
Default design for DTH11 Temp/Humidity sensor.
J20
I2C_Temp_Conn
+3.3V
SDA_GPIO02
SCL_GPIO03
GND
Alternative I2C Pin layout for future use.
Mücahit Kaya
AG Schumann
Sheet: /
File: Board_Desing_V4.kicad_sch
Title: Muon_Telescope_Control_Board
Size: A4
Date: 2026-06-30
Rev: v3
KiCad E.D.A. 10.0.6
Id: 1/1